GEK-83856

DRIVE SYSTEM INSTRUCTIONS

AC GENERATOR/ENGINE' MODULE (GEM)

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL@ ELECTRIC

AC GENERATOR/ENGINE MODULE (GEM)

(GEK - 83856)

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DS3820WGC __*__ AC GENERATOR/DIESEL ENGINE

VOLTAGE REGULATION AND ENGINE GOVERNOR MODULE

*See actual module for B, C, D or E version module

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GEK-83856

NOTICE

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, 1501 Roanoke Boulevard, Salem, Virginia, USA, 24153.

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7

SAFETY SYMBOL LEGEND



Commands attention to an operating procedure; practice; condition or statement, which, if not strictly observed, could result in personal injury or death.



Commands attention to an operating procedure; practice; condition or statement, which; if not strictly observed; could result in damage to, or destruction of equipment;

NOTE

Commands attention to an essential operating or maintenance procedure, condition or statement which must be highlighted.

TABLE OF CONTENTS

DESCRIPTION	PAGE NUMBER
FUNCTION	1
THEORY OF OPERATION	1
GOVERNOR CONTROL	1
VOLTAGE REGULATION	1
METERING	2
FAULT DETECTION	2
FLOW DIAGRAM INTERPRETATION	4
Flow Diagram "A" Operation of GEM Governor	6
Flow Diagram "B" Operation of GEM Voltage Regulator and Power Supply	7
Flow Diagram "C" Operation of GEM Metering	8
Flow Diagram "D" Operation of GEM Fault Circuitry	9
GEM ELEMENTARY DIAGRAMS (173C5527, pages 4A through 4D)	10
REMOVAL AND REPLACEMENT PROCEDURES	15
AC GENERATOR CONTROL MODULE INTERCONNECTIONS (DS3820WGC, sheet 4AA)	16

NOTES

FUNCTION

The AC Generator Engine Module (GEM) is used to:

1. maintain constant engine speed,
2. regulate the generator output voltage
3. Provide fault diagnostics

THEORY OF OPERATION

GEM operation is divided into four functional areas:

1. governor control,
2. voltage regulation,
3. metering, and
4. fault detection.

Refer to the flow diagrams while reading the next four functional descriptions;

A flow diagram is a condensed, easy to read summary of the operation of a circuit or module. If you are not familiar with the symbols used in flow diagrams, be sure to read the section titled "Flow Diagrams",

GOVERNOR CONTROL

The GEM governs the engine speed by controlling the engine actuator (throttle) position with an electrical signal. The Engine Control Selector on the front of the GEM is involved in engine speed control and has four positions:

1. OFF
2. IDLE
3. RUN, and
4. LOAD

In the OFF position, the engine is shut down. In the IDLE position the engine can be started and will run at idle speed. In the RUN position, the engine runs at normal speed but the generator will not pick up or share load with the other on line generators. In the LOAD position the engine runs at normal speed and the generator will pick up load and work with other on line generators to equally share the load.

The Frequency Trim Adjustment located below the Engine Control Selector is used to match the speed of the generator being brought on line to the AC through bus frequency.

VOLTAGE REGULATION

The GEM will regulate generator output voltage to a Steady value under changing loads. This function is implemented by comparing the generator output voltage with a reference and increasing or decreasing the generator field voltage.

METERING

There are four external meters that monitor output:

1. Kilowatt Meter - Indicates how much power is being generated by the engine. Used to ensure that the engines are all sharing the load.

2. Kilovar Meter - Indicates the reactive power delivered by the generator; Used to ensure that the generators are properly paralleled.

3. Voltmeter - Indicates the voltage of the source phases selected by the four position Voltmeter Switch on the GEM front panel. The first three positions of the switch select the line-to-line voltage of the generator to be monitored. The fourth position selects the AC bus voltage to be monitored. **(VL1-2; VL2-3; VL3-1; VBus)**

4. Ammeter - Indicates the line current for the phase selected by the Amps Switch on the GEM front panel.

FAULT DETECTION

One red and six yellow indicating lights on the front panel of the GEM annunciate fault conditions. Under normal operating conditions with the generator on line, all lights (yellow and red) will be on. The first fault to occur causes its associated light to flash and subsequent faults cause the associated lights to go out. Pressing the reset pushbutton will reset the control, but if a fault still exists, the appropriate light will flash and/or go out. All faults must be cleared before the generator breaker can be closed. The fault lights are listed and described on the following page.

FAULT DETECTION (Continued)

1. Gen On Light - When ON, indicates that the engine/generator is running, the breaker is closed, and power is being supplied to the AC bus. When OFF, indicates that the breaker is open.

2. Undervolt Light - When ON, indicates a generator voltage greater than 75% rated voltage. When OFF, indicates that a generator undervoltage condition exists. This condition causes the generator circuit breaker to trip.

3. No OV/UF Light - When ON, indicates a normal condition. When OFF, indicates that a generator overvoltage or underfrequency condition exists. The generator circuit breaker WILL TRIP FOR AN OVERVOLTAGE CONDITION, BUT NOT FOR AN UNDERFREQUENCY CONDITION.

4. No Rev Pwr Light - When ON, indicates a normal condition. When OFF; indicates that reverse power was flowing from the AC bus back to the generator. This condition causes the generator circuit breaker to trip.

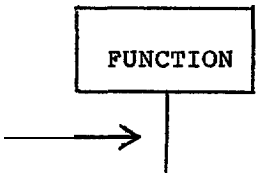
5. No Ovrspd Light - When ON, indicates a normal condition. When OFF, indicates that the generator engine was overspeeding. Overspeeding causes fuel to be cut off from the engine and the generator breakers to open. The engine should be supplied with mechanical overspeed protection.

6. No Tach Loss Light - When ON, indicates a normal condition. When OFF, indicates that the tachometer signal has been lost. This condition causes the generator engine to shut down and the generator breakers to open.

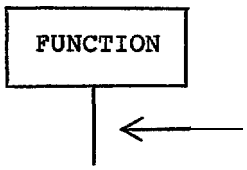
7. Pwr Supply OK Light - When ON, indicates a normal condition. When OFF, indicates that the generator control power supply has malfunctioned. This condition causes the generator engine to shut down and the generator breakers to open.

FLOW DIAGRAMS

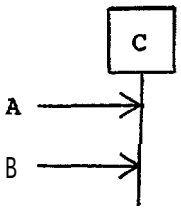
The signal flow diagrams on the following pages use the following symbols to help explain each function:



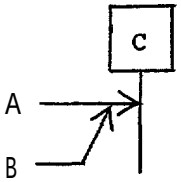
A signal terminating on the left of the center line is always a permissive to start and maintain the function once it is started. If any one of these permissives is lost, the function becomes inactive.



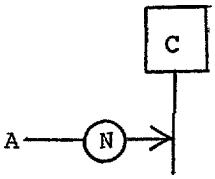
A signal terminating on the right of the center line is required only at the start of the function. After starting, if the signal is lost the function remains active.



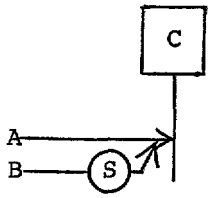
This is a logical AND function.
A • B = C
A and B cause function C.



This is a logical OR function.
A + B = C
A or B causes function C.



This is a logical NOT function.
$\overline{A}$ = C
If signal A is not present, then function C is active.



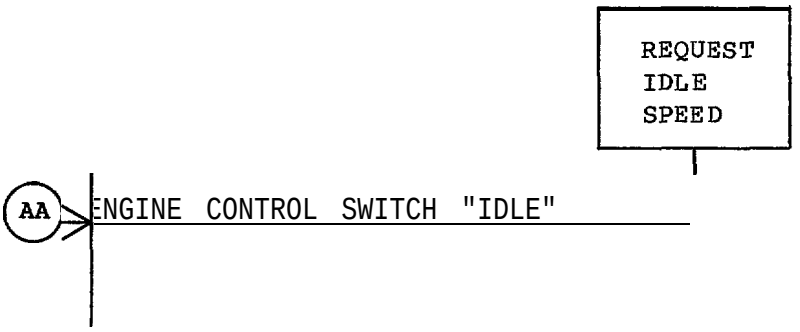
This is a conditioned "OR" or "seal in" function.
A = C
A causes function C, which is then "sealed" by B.

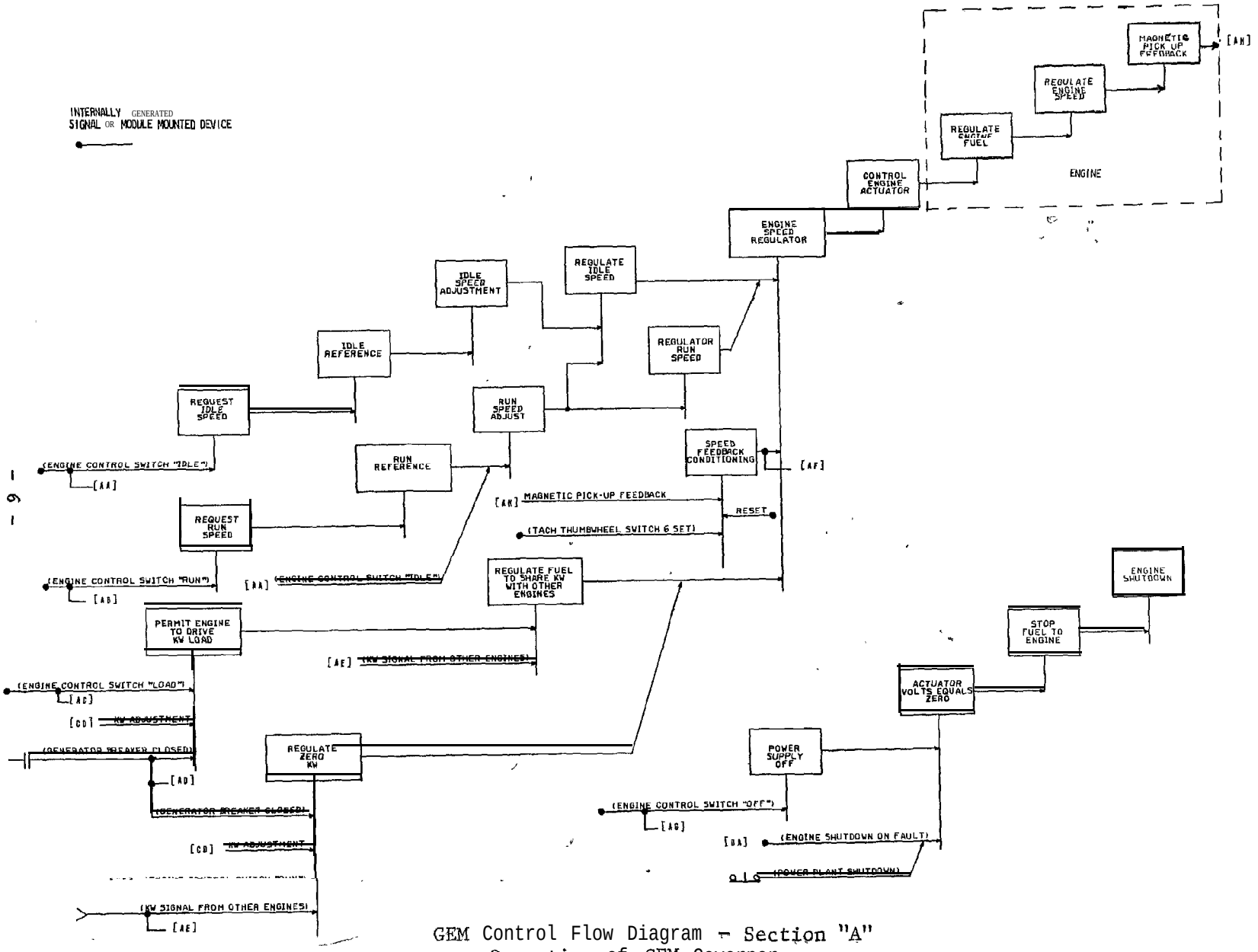


This is a time delay function.

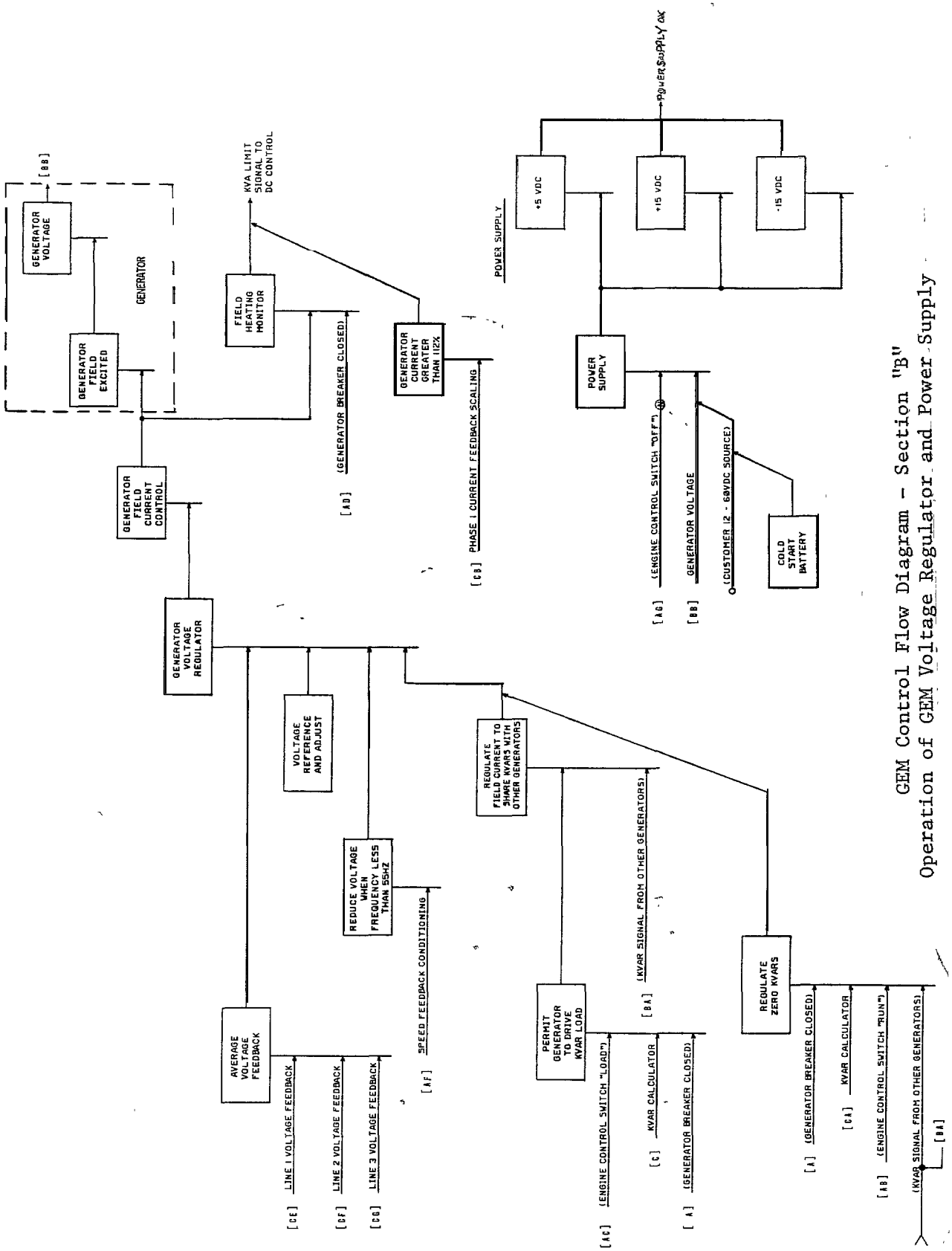
FLOW DIAGRAMS (Continued)

Signals are cross referenced from sheet to sheet by being labelled with a sheet identification (A, B, C, or D, which specifies on which sheet the signal originates) and a label (A, B, C, etc., which differentiates between several several signals on a sheet) within brackets. In the following example, the signal ENGINE CONTROL SWITCH "IDLE" is the first one used on sheet A, hence its identification as AA. The next signal used would be AB, and so forth. This signal is shared with other points cross referenced in the circle or brackets. If more than one point shows AA, it only means that all those points share the same signal.

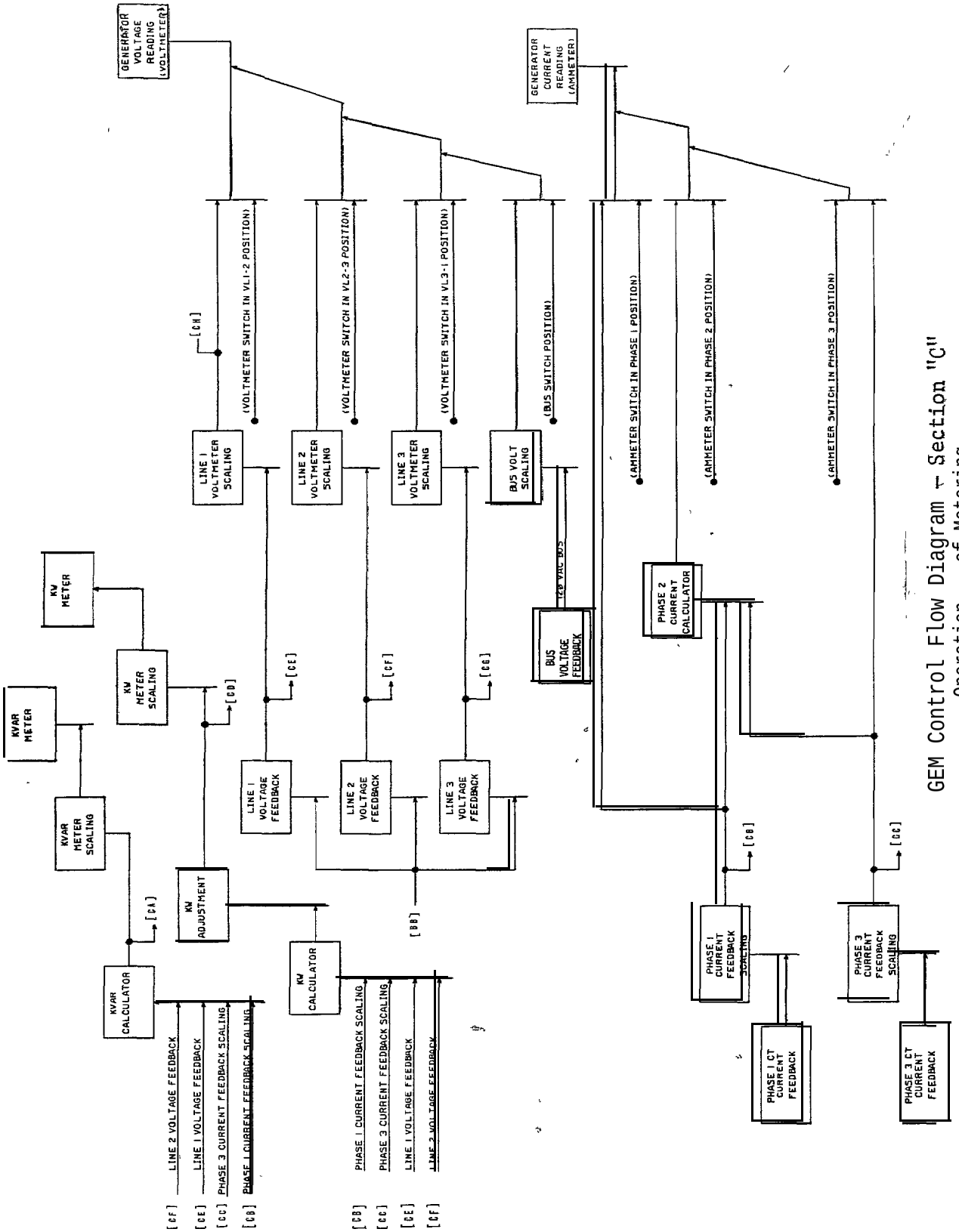




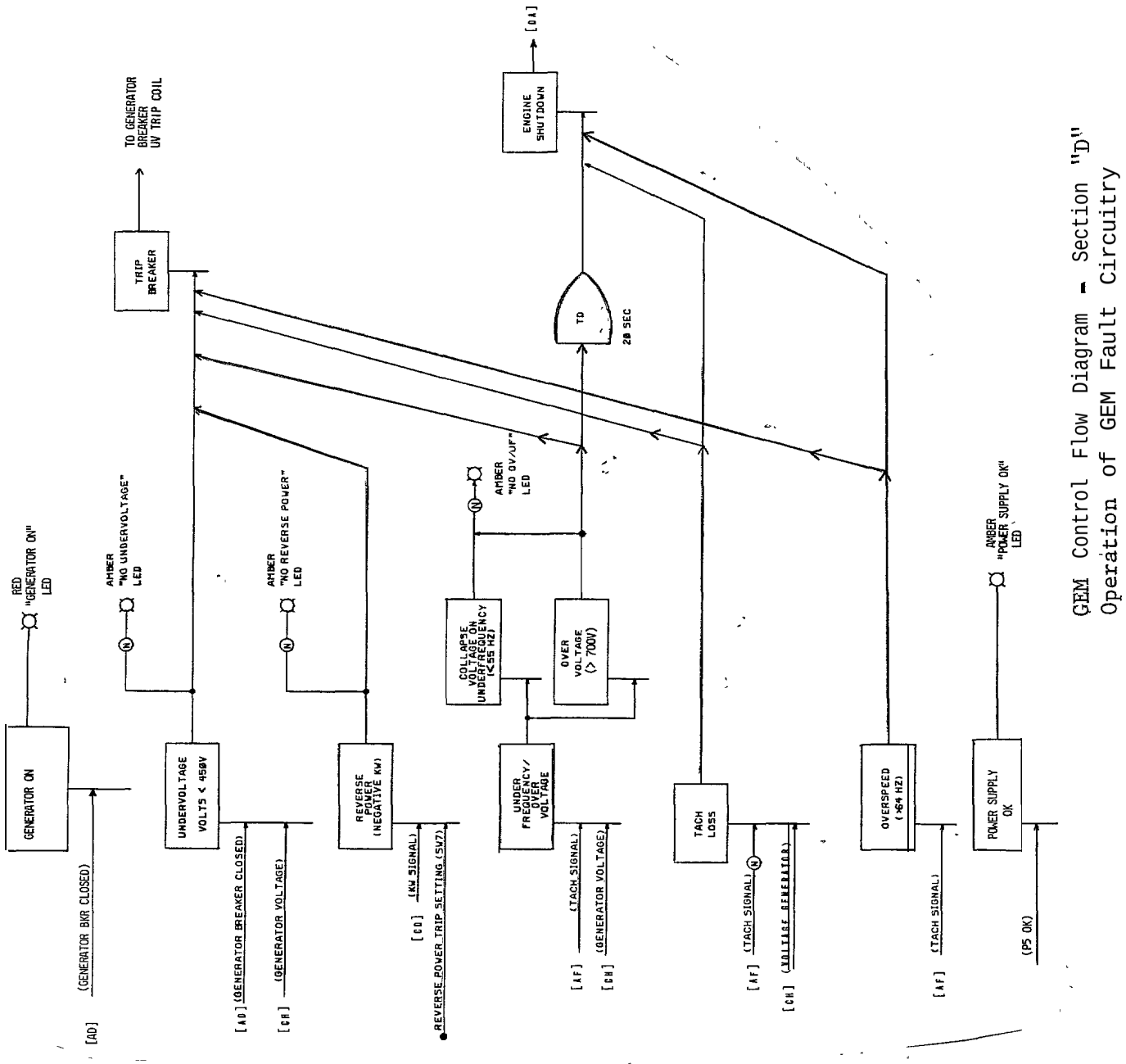
GEM Control Flow Diagram – Section "A"
Operation of GEM Governor



GEM Control Flow Diagram - Section "B"
Operation of GEM Voltage Regulator and Power Supply



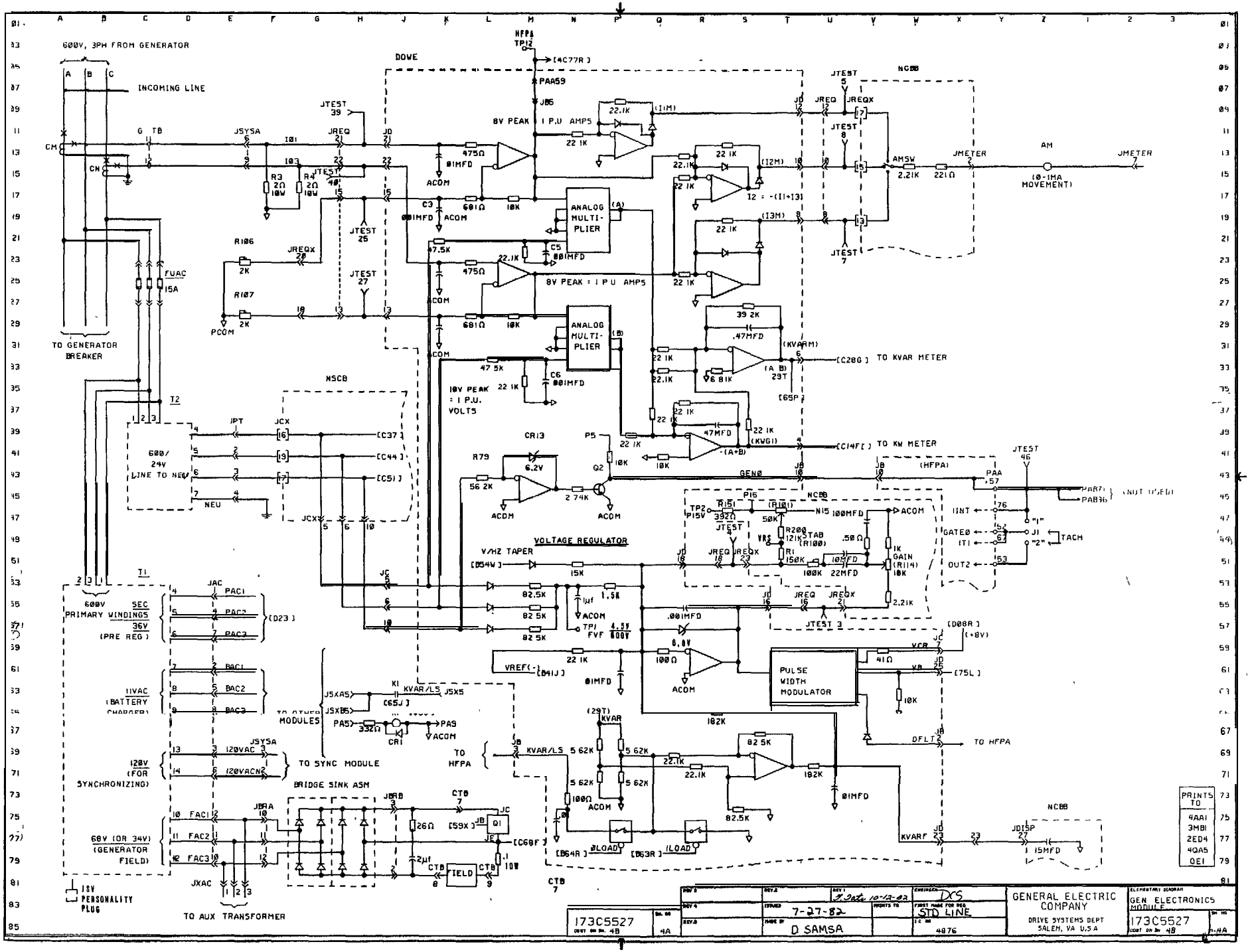
GEM Control Flow Diagram - Section "C"
Operation of-Metering.

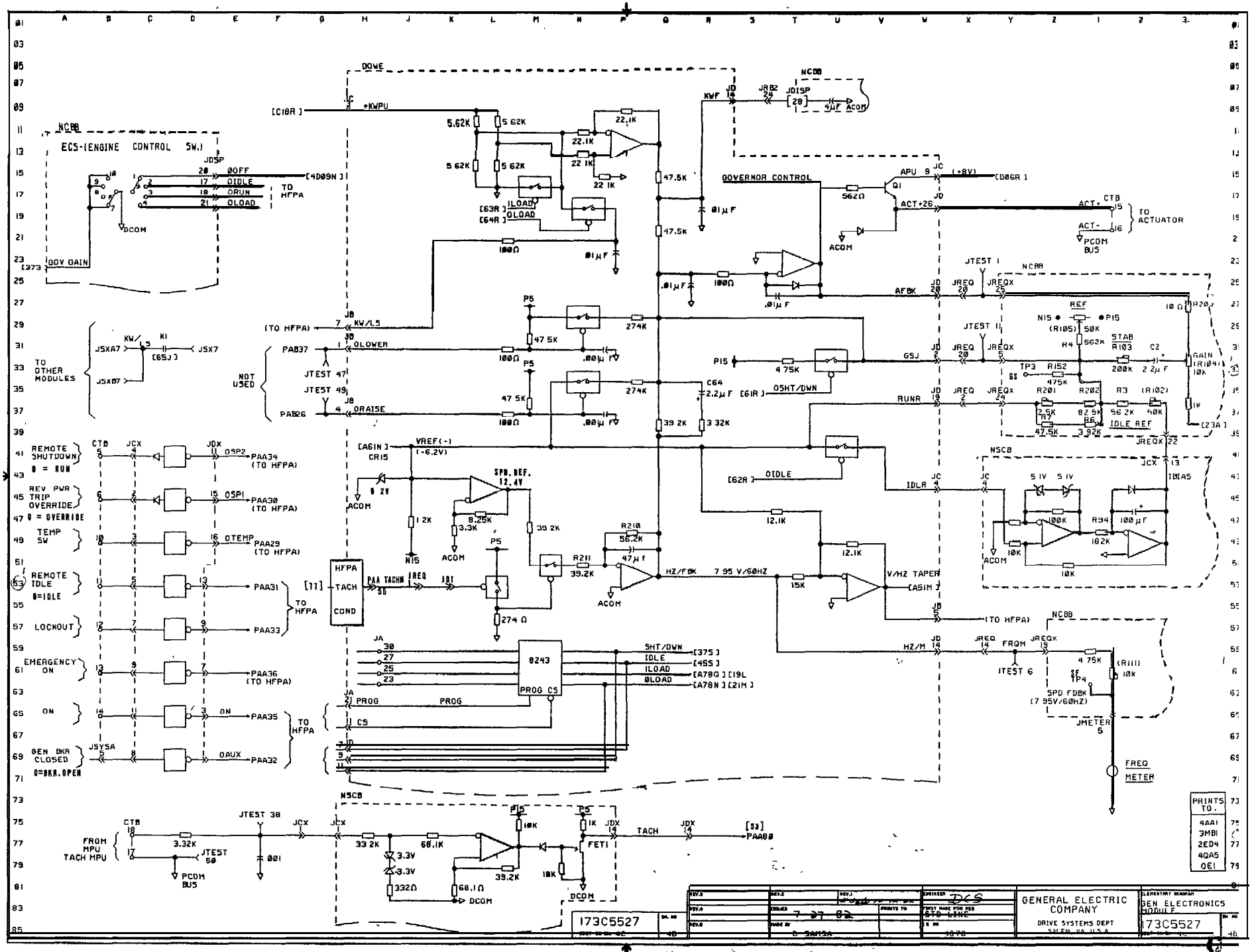


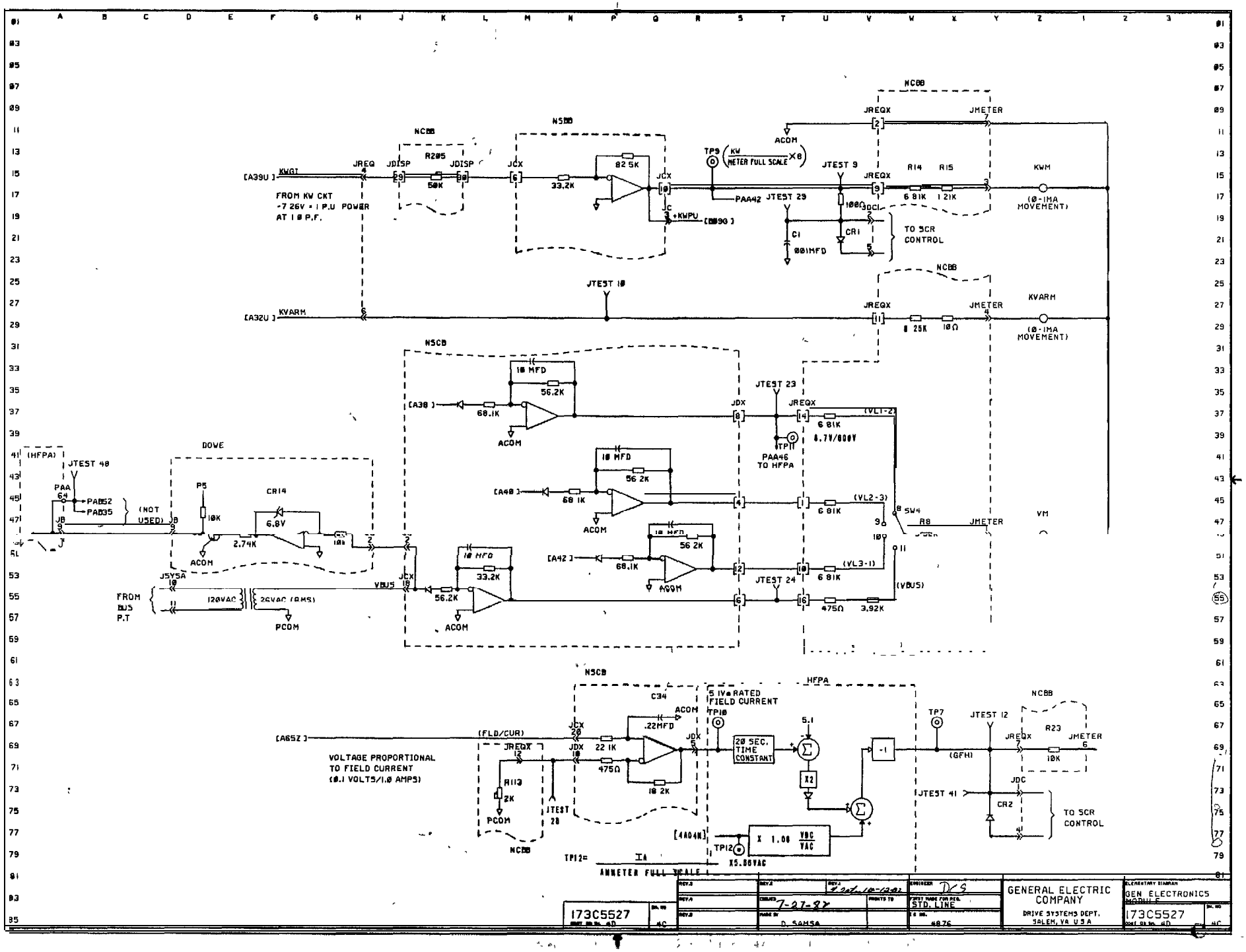
GEM Control Flow Diagram - Section "D"
Operation of GEM Fault Circuitry

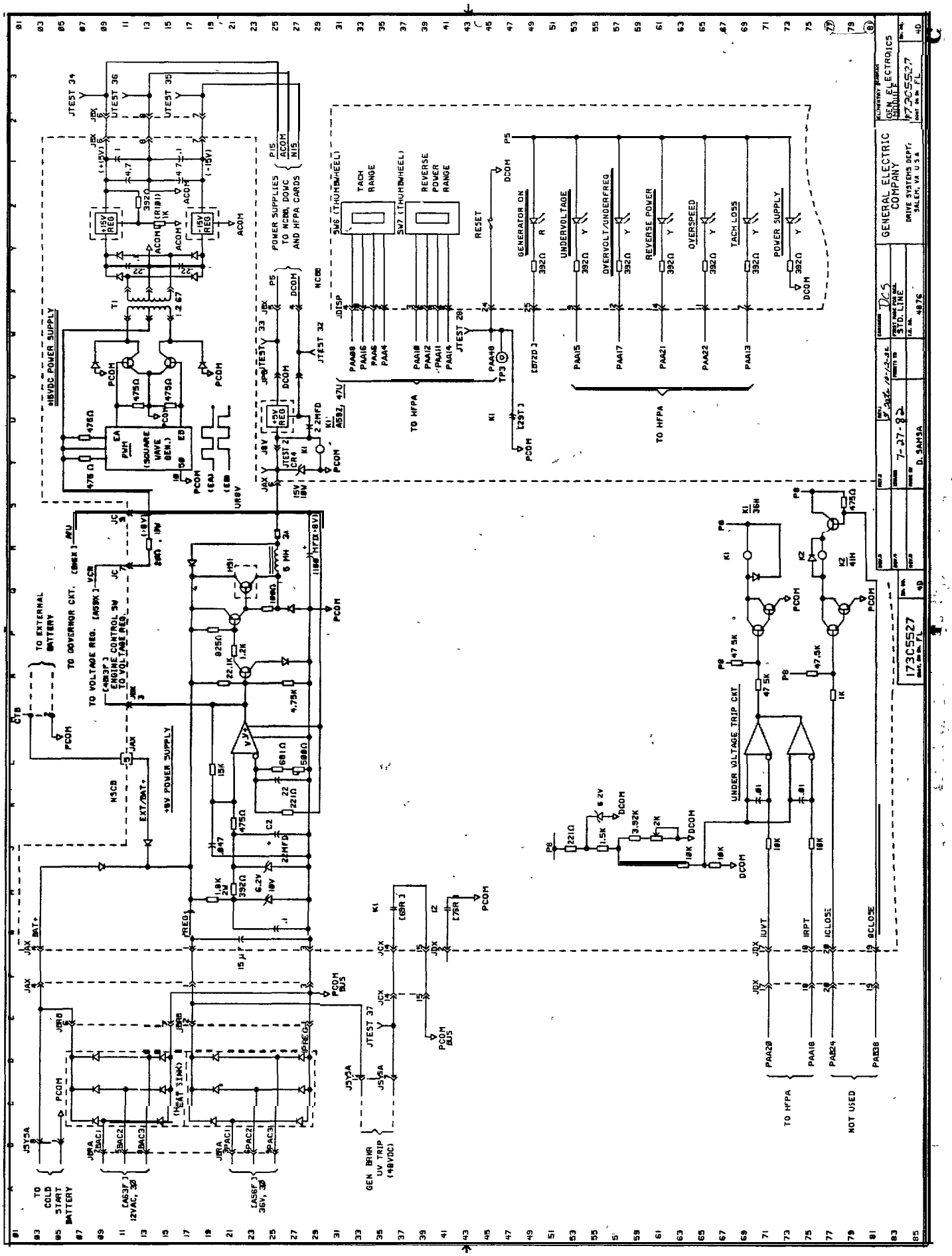
ELEMENTARY DIAGRAMS

Sheets 4A through 4D contain elementary diagrams for the AC generator/engine module.


600V, 3PH FROM GENERATOR
INCOMING LINE
TO GENERATOR BREAKER
DOWE
NSCB
NCBB
ANALOG MULTI-PLIER
8V PEAK = 1 P.U. AMPS
10V PEAK = 1 P.U. VOLTS
TO KVAR METER
TO KW METER
TO AM (0-1MA MOVEMENT)
VOLTAGE REGULATOR
V/HZ TAPER
PULSE WIDTH MODULATOR
600V SEC PRIMARY WINDINGS
36V (PRE REG)
110VAC (BATTERY CHARGER)
120V (FOR SYNCHRONIZING)
68V (OR 34V) (GENERATOR FIELD)
TO SYNC MODULE
TO OTHER MODULES
TO HFPA
TO AUX TRANSFORMER
BRIDGE SINK ASM
ISV PERSONALITY PLUG
600/24V LINE TO NEU
GEN ELECTRONICS MODULE
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA U.S.A.
173C5527
D SAMSA
7-27-82
STD LINE


ECS-(ENGINE CONTROL SW.)
GOVERNOR CONTROL
REMOTE SHUTDOWN 0 = RUN
REV PWR TRIP OVERRIDE 0 = OVERRIDE
TEMP SW
REMOTE IDLE 0=IDLE
LOCKOUT
EMERGENCY ON
ON
GEN BKR CLOSED 0=BKR.OPEN
TO OTHER MODULES
NOT USED
TO HFPA
TO ACTUATOR
SPD. REF. 12.4V
VREF(-) (-6.2V)
8243
PROG CS
FROM MPU TACH MPU
SPD FDBK (7.95V/60HZ)
FREQ METER
GEN ELECTRONICS MODULE
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT
SALEM, VA. U.S.A.
173C5527


NCBB
NSBB
JREQX
JMETER
ACOM
KWGT
[A39U]
JREQ
JDISP
R205
50K
33.2K
82.5K
JDX
JCX
TP9
KW METER FULL SCALE X8
PAA42
JTEST 9
R14
R15
6.81K
1.21K
KWM
(Ø-1MA MOVEMENT)
FROM KW CKT -7.26V = 1 P.U POWER AT 1.0 P.F.
JTEST 29
100Ω
JDC
C1
.001MFD
CR1
TO SCR CONTROL
JTEST 10
KVARM
[A32U]
8.25K
10Ω
NSCB
10 MFD
56.2K
[A38]
68.1K
JTEST 23
(VL1-2)
8.7V/600V
TP11
PAA46 TO HFPA
[A48]
56.2K
(VL2-3)
SW4
R8
VM
[A42]
(VL3-1)
JTEST 24
(VBUS)
475Ω
3.92K
DOVE
(HFPA)
JTEST 48
PAA 64
PAB52
PAB35
(NOT USED)
P5
10K
CR14
6.8V
2.74K
J5Y5A
FROM BUS P.T
120VAC
26VAC (RMS)
PCOM
VBUS
33.2K
56.2K
[A65Z]
(FLD/CUR)
VOLTAGE PROPORTIONAL TO FIELD CURRENT (0.1 VOLTS/1.0 AMPS)
HFPA
C34
.22MFD
22.1K
475Ω
10.2K
R113
2K
JTEST 28
5.1V RATED FIELD CURRENT
TP10
20 SEC. TIME CONSTANT
5.1
X2
-1
[4A04K]
X 1.00 VDC/VAC
TP12
X5.86VAC
TP12= 1A AMMETER FULL SCALE
TP7
JTEST 12
R23
10K
(GFH)
JTEST 41
CR2
TO SCR CONTROL
173C5527
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA U.S.A.
STD. LINE
7-27-82
D. SAMSA
GEN ELECTRONICS MODULE
173C5527


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SALEM, VA U.S.A.
GEN. ELECTRONICS MODULE
173C5527
STD. LINE
7-27-82
D. SAMRA
TO EXTERNAL BATTERY
TO GOVERNOR CKT.
TO VOLTAGE REG.
ENGINE CONTROL SW
TO VOLTAGE REG.
+8V POWER SUPPLY
EXT/BAT+
NSCB
PCOM
DCOM
UNDER VOLTAGE TRIP CKT
TO HFPA
NOT USED
GEN BRKR UV TRIP (48VDC)
TO COLD START BATTERY
POWER SUPPLIES TO NCBB, DOWC AND HFPA CARDS
TACH RANGE
REVERSE POWER RANGE
RESET
GENERATOR ON
UNDERVOLTAGE
OVERVOLT/UNDERFREQ
REVERSE POWER
OVERSPEED
TACH LOSS
POWER SUPPLY
+15VDC POWER SUPPLY
PWM (SQUARE WAVE GEN.)

REMOVAL AND REPLACEMENT PROCEDURES

Generator Control Module (GEMA)

REMOVAL:

1. Disconnect all power to the panel; Note the location of plugs to be disconnected: JXAC, JSYSA, JSYSB, JSYSC, JPT, JDC1, JDC2, and JDC3. Also note the order of connections at the GEMC terminal board at the top of the module. Disconnect all of these wires and plugs.

2. Loosen the 6 GEMA mounting screws (three on each side of the module) two full turns each. Grasp the module at the steel sides covering the card rack. Lift upward and toward the front of the panel to withdraw the mounting screws through the teardrop mounting holes. Remove the module from the panel.

REPLACEMENT: Mount the replacement module by reversing the order of the above steps.

Generator Control Transformer Module (GEMB)

REMOVAL:

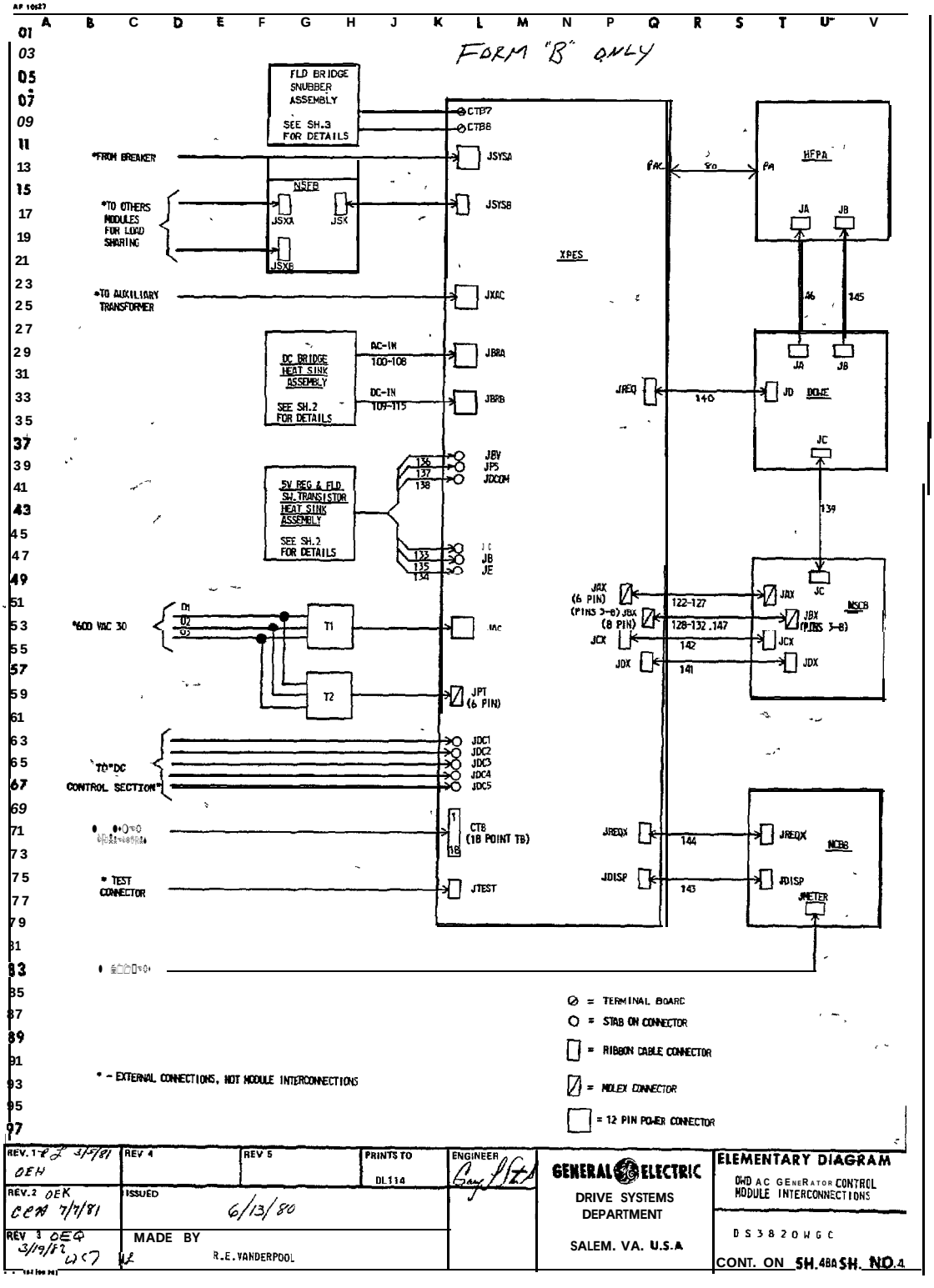
1. Disconnect all power to the panel. Remove the GEMA module according to the removal instructions above.

2. Loosen the four GEMB mounting screws (two at the top and two at the bottom) two full turns each. Firmly grasp the framework and lift upward and toward the front of the panel to withdraw the mounting screws through the teardrop holes at the top of the module and the slots at the bottom. Remove the module from the panel and place it upright on a table beside the replacement module.

REPLACEMENT:

1. Facing the rear of the modules, ensure that the jumpers on the transformer terminal board are connected exactly the same on the replacement module as on the original module. Later versions of this module have a nameplate located on the sideplate which identifies the jumper connections for a given generator field voltage.

2. Replace the module by reversing the order of the above removal procedure.


FORM "B" ONLY
FLD BRIDGE SNUBBER ASSEMBLY
SEE SH.3 FOR DETAILS
CTB7
CTB8
*FROM BREAKER
JSYSA
NSEB
*TO OTHERS MODULES FOR LOAD SHARING
JSXA
JSX
JSXB
JSYSB
XPES
*TO AUXILIARY TRANSFORMER
JXAC
DC BRIDGE HEAT SINK ASSEMBLY
SEE SH.2 FOR DETAILS
AC-IN 100-108
DC-IN 109-115
JBRA
JBRB
5V REG & FLD SH. TRANSISTOR HEAT SINK ASSEMBLY
SEE SH.2 FOR DETAILS
136
137
138
JBV
JP5
JDCOM
133
135
134
JB
JE
*600 VAC 3Ø
T1
T2
JAC
JPT (6 PIN)
TO "DC CONTROL SECTION"
JDC1
JDC2
JDC3
JDC4
JDC5
CTB (18 POINT TB)
1
18
* TEST CONNECTOR
JTEST
HFPA
JA
JB
FA
FAC
50
A6
145
JREQ
140
JD
DCME
JC
139
JAX (6 PIN)
JBX (8 PIN)
JCX
JDX
122-127
128-132,147
142
141
NSCB
(PINS 3-8)
JREQX
JDISP
144
143
MCB6
JMETER
= TERMINAL BOARD
= STAB ON CONNECTOR
= RIBBON CABLE CONNECTOR
= MOLEX CONNECTOR
= 12 PIN POWER CONNECTOR
* - EXTERNAL CONNECTIONS, NOT MODULE INTERCONNECTIONS
REV. 1 OEH 3/9/81
REV 4
REV 5
PRINTS TO DL114
ENGINEER
REV.2 OEK CRM 7/7/81
ISSUED 6/13/80
REV 3 OEH 3/19/81 WCT
MADE BY R.E. VANDERPOOL
GENERAL ELECTRIC
DRIVE SYSTEMS DEPARTMENT
SALEM. VA. U.S.A
ELEMENTARY DIAGRAM
DWD AC GENERATOR CONTROL MODULE INTERCONNECTIONS
DS3820WGC
CONT. ON SH.4BA SH. NO.4

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